Building a Compass to Explore the Cryptoverse

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1. 1. Introduction: Understanding the Crypto Phenomenon Beyond Appearances

Since their creation in 2009, with Bitcoin, cryptocurrencies have experienced a turbulent yet rich history. Their adoption has exploded, growing from a few thousand to nearly 900 million users (Statista). However, despite this spectacular growth, they remain poorly understood by the public.

a. A Divided Ecosystem: Between Memes and Real Utility

This lack of understanding is partly explained by the dominant image of cryptocurrencies, often associated with instability, pure speculation, meme-trading, or the persistent regret of not having invested in Bitcoin in its early days.

Yet behind this noisy facade, an underlying movement is underway. As revealed by "The 2025 Geography of Crypto Report", cryptocurrencies are seeing growing adoption, particularly in developing regions such as Asia, Latin America, or Africa.

This adoption is based on pragmatic needs:

- Protection tool against inflation and currency devaluation (Nigeria, Turkey for example).
- Daily usage as a means of payment or for international transfers (Vietnam, Nigeria for example).
- Conversely, it is in the most developed economies (Europe, North America) that the speculative role of crypto remains the most marked.

Simultaneously, structural innovations are redefining the ecosystem:

- Institutional Adoption: Approval of ETFs, institutional adoption in the United States, or Tokenization of Real-World Assets (RWA).
- Decentralized Infrastructure: New storage solutions (like Filecoin), decentralized computing (Akash for GPU rental, Render Network used in 3D project rendering), decentralized AI training with Bittensor, or bandwidth rental with GRASS.

Faced with this duality, speculation on one side and concrete adoption on the other, it becomes mandatory to adopt an objective analysis framework. It is to navigate with a clear direction in this complex universe that I initiated this data science project, to equip myself with a rigorous exploration and analysis tool.

b. First Step: From Intuition to Method

My approach to the ecosystem relies above all on the need to establish a rigorous method.

As a former Project Manager / Business Developer, accustomed to multicriteria optimization, and today a Data Scientist specialized in evaluating artificial intelligence models and data quality, it is impossible for me to approach the crypto universe through its purely psychological angle.

I speak of psychology because it is an important component in finance. But, in crypto, it is amplified by social networks, a source of FUD/FOMO (Fear, Uncertainty, Doubt / Fear of Missing Out).

In science, we often start, initially, by simplifying the problem. Here, this means focusing on price alone, to see if it contains an exploitable signal.

I therefore naturally began to focus my interest on Bitcoin, both the starting point and the true lung of the ecosystem.

For this reason, I chose to build an objective analytical approach, based on data, to question what fundamentally makes the value of a project. Especially since one of the promises of cryptocurrencies is that of transparency and accessibility of this data.

c. A Logbook as a Thinking Laboratory

To carry out this exploration, I chose to structure my entire approach into a series of four articles, in different formats, a sort of logbook that aims to be transparent and methodological.

This journey will be articulated as follows:

- This article: Presentation of the approach and objectives.
- Article 2 (Notebook): The Foundations: Mastering Data Source.
- Article 3 (Notebook): The Core of the Experiment: ML Modeling.
- Article 5: A New Direction.

2. Starting Postulate: Can We Predict Tomorrow Using Yesterday?

a. The Tested Hypothesis: Can We Predict Price with Price Alone?

The idea here is to know if past price alone is sufficient to determine future price. Because by reading on the internet, it is possible to find both answers to this question. It is one of the most heated debates in quantitative finance. It is therefore a question of forming my own opinion on the matter, based on my own experiences and a reproducible methodology, with all my codes being public.

Time series in finance are often considered an extremely complicated exercise. They are said to be non-stationary and potentially random walks, which means that yesterday's signals are not good predictors for tomorrow.

The central question is: with the recent technological leap in Artificial Intelligence (notably recurrent networks like LSTM and GRU that I will use), is it possible to solve this problem or, at least, extract an exploitable signal that would contradict the market inefficiency hypothesis?

b. A Three-Step Approach: Data, Exploration, Modeling

The first step to test this hypothesis is to establish an action plan. For this, I informed myself upstream by reading articles, GitHub repos, and Kaggle notebooks. This quick review allowed me to confirm a clear direction, and to decide on a three-step plan:

- The Data Pipeline: Any modeling project without reliable and quality data is doomed to failure. In addition to guaranteeing data integrity, seeking to master the acquisition of a data retrieval system will allow us to integrate other cryptos more easily later.
- Exploratory Analysis (EDA): Once the dataset is retrieved, the next step is to ensure its quality, and to understand the data oneself. EDA is the indispensable basis for extracting relevant information and deciphering market dynamics.
- Modeling: Finally, it will be time to establish artificial intelligence models. We will start by establishing a base model (inspired by online codes), to ensure the feasibility of the approach, before moving to an optimization step to obtain the best possible results.

It is at the end of this approach that we will be able to determine if the predictability hypothesis by price alone is validated or, on the contrary, rejected.

3. The Roadmap : Build Before Predicting

The starting hypothesis being established, as well as the necessity of having a rigorous approach, we can detail the roadmap.

In the following parts, we will present above all the philosophy, and the major issues of each of the planned approaches. Technical aspects, including code, detailed analyses, and raw results, will be available in the associated notebooks.

Finally, particular attention was paid to the modularity of the code, for several reasons:

- To be able to reuse it for other cryptos easily, or in other projects.
- To train myself to write clean, readable, and maintainable code.
- To improve the overall readability of the project.

a. Article 2 – The Foundations: Mastering the Data Source

As mentioned previously, data quality is the number one success factor in Machine Learning. It is impossible to obtain reliable results when data is incomplete, biased, or poorly understood.

Building a Data Pipeline

It is possible to download "ready-made" datasets, but these present several disadvantages:

- Obsolescence: They are frozen, and therefore do not benefit from real-time updates.
- Quality: Depending on their source, their acquisition and cleaning method, their quality is not assured.
- Flexibility: They are not necessarily available for all cryptocurrencies.

Another option would be on-chain analysis, that is, querying the blockchain directly to obtain information. But once again, this presents different technical disadvantages:

- Reduced Modularity: A method must be developed for each blockchain, which increases complexity, and reduces modularity.
- Data Limitation: Blockchain protocols record information including transactions, but not the prices of amounts exchanged.

To obtain this data efficiently, it is therefore preferable to go through a market API. Although these services are often paid for historical data, the Binance platform offers the possibility to obtain them for free, at the cost of some technical work. Indeed, it only allows retrieving data in packets of 1000 lines. Despite this small technical challenge, it is possible to develop an automated solution to overcome this limitation.

Exploratory Data Analysis (EDA)

Now that a reliable data retrieval method has been developed, we must ensure that the data is reliable and usable.

Initially, we use a classic approach: we check for the presence of missing or aberrant values, thanks to functionalities provided by data science libraries like Pandas. Plotting graphs is also essential to visualize data continuity.

In model training, we will only use closing prices. However, the downloaded data contains much more information (Volume in BTC, Number of Transactions, etc.). It is therefore relevant to extract information from it.

This is where I chose to deviate from the traditional approach. Rather than turning to classic technical indicators from finance (RSI, Bollinger Bands or others), which are widely available online and on which I would have no added value, I sought to bring my own analysis by focusing on:

- Market Behavior: Analyzing the distribution of average transaction size to identify if the Binance market is dominated by retail or institutions.
- Internal Dynamics: Studying buyer/seller pressure and volumes to understand the internal dynamics of exchanges.
- Anomalies: Identifying trend or cycle breaks that could signal a deep change in the market (like those observed in 2025).

It is from this exploratory analysis that the decision to limit the model training period to the year 2025 only was made.

Why? Because we were able to highlight several distinct market phases, with different actor behaviors. Training the model on a period where market dynamics and behaviors are heterogeneous would skew the results obtained.

Thus, EDA reveals its importance: it should not be perceived as an optional step, but as an essential basis for modeling.

b. Article 3: The Core of the Experiment: ML Modeling

To establish an initial reference and validate the feasibility of the project on Bitcoin data, we started with a simple implementation, inspired by existing tutorials. This approach quickly highlighted a major methodological limitation.

Most time series implementations found online use a so-called "Single-step" approach (or step-by-step prediction), often directly provided by machine learning libraries.

The principle is simple, the model takes a window of historical data as input, then predicts a single future value. Then, the window slides one step in time, replacing the oldest data with the actual observed value, and not with the predicted value. This scheme is illustrated in the following figure.

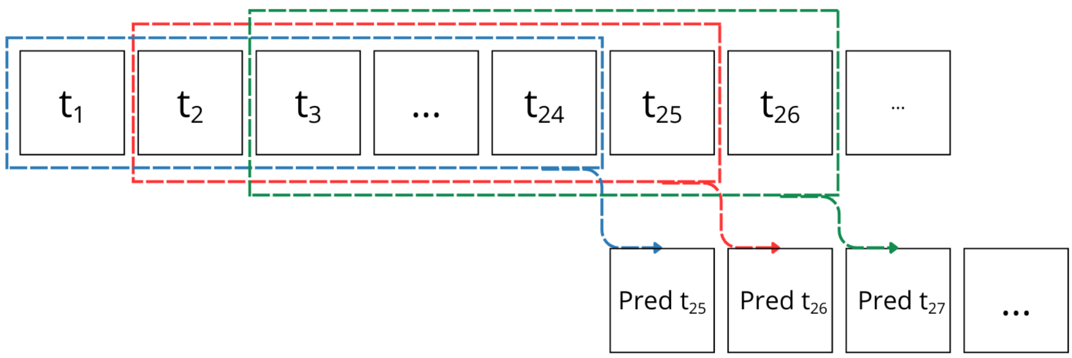

Figure 1. Explanatory diagram of the single-step approach

Although this method often generates beautiful metrics (the model being constantly "corrected" by reality), it presents little practical interest. If one asks what the price of Bitcoin will be in one hour (the time step used), one takes little risk in answering that it will still be roughly the same. No need for artificial intelligence to guess that.

Furthermore, this approach is inapplicable in production:

- It assumes continuous and immediate access to the actual value at the time of prediction, which requires a data pipeline kept up to date permanently, which can lead to significant additional costs.
- Worse still: if the temporal granularity is too short (to the second or even lower in high-frequency trading), and the time necessary to retrieve the data, prepare the

model input and produce the prediction is greater than this time step, then the result is obsolete before even being calculated.

It is to try to stick as closely as possible to operational reality that an alternative method was implemented, inspired by the official TensorFlow documentation.

This is a "multi-step" approach, in which the generated predictions are reinjected as input to feed the following predictions. Thus, the model no longer depends on future real values, only on its own predictions as illustrated in the following figure:

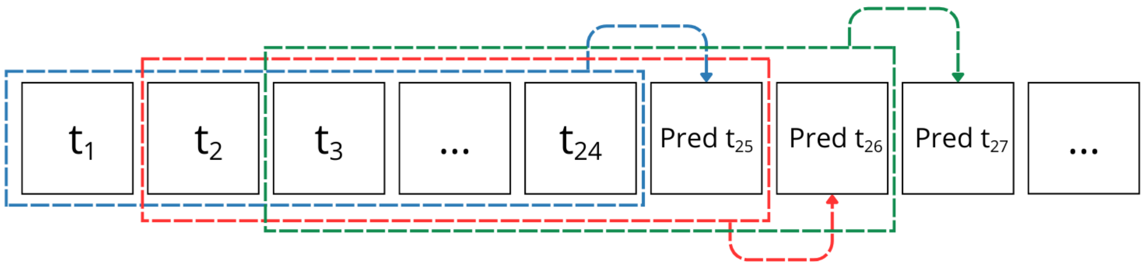

Figure 2. Explanatory diagram of the multi-step / Walk forward approach

This is where the true value of this work lies: testing models in conditions close to operational reality.

All that remains is to implement this and thus decide on the tool to use among the array available to the Data Scientist.

Recurrent Neural Networks (RNN) impose themselves naturally: they are specialized in processing sequences, that is, ordered data, whether text or, in our case, time series.

Three architectures were selected:

- SimpleRNN, the most basic model, but limited by its short-term memory.
- LSTM (Long Short-Term Memory), an established standard in data science and finance, capable of retaining long-term dependencies, at the cost of greater resource consumption.
- GRU (Gated Recurrent Unit), a more recent alternative, designed to reduce complexity while retaining a learning capacity close to that of LSTM.

The SimpleRNN model serves as a baseline, as a reference, LSTM as an industrial benchmark, and GRU as an optimized alternative.

In parallel, several hyperparameters were selected for the initial experimentation (their technical details are reserved for the notebooks).

To set up this process, two important elements guided the design:

- Modular Code: to facilitate modification of hyperparameters and management of multiple tested configurations.
- Traceability: because the growing volume of experiments makes precise tracking of results indispensable. This is why MLflow was integrated from the start.

Following these experiments, it is possible to draw several results:

- Although it is possible to maintain relatively stable predictions over a few iterations, errors accumulate rapidly (error compounding phenomenon), which quickly distances predictions from reality.
- Regardless of the level of hyperparameter optimization, no model manages to capture the fine variability of prices. They only learn a general trend, often inconsistent from one experiment to another.

To eliminate any disturbing influence of the overall price trend (increase or decrease over the period), we introduced log returns (logarithmic returns). This transformation makes the series stationary, that is, centered around a zero mean, and focuses on relative variations rather than absolute levels.

The conclusion is then unequivocal, the curve predicted by the model is a line at zero, it is the demonstration that the latter learns nothing from the data provided to it.

The most obvious conclusion is therefore that the initial hypothesis is false: there is not enough information in past prices alone to predict future prices.

c. Article 4: A New Direction

The next step is to determine the reason or reasons why the initial hypothesis is invalidated.

The analysis of results obtained in Article 3 revealed several blocking points that can be summarized as follows: price is an output variable, resulting from a multitude of external factors invisible in a simple time series. Attempting to predict price with price alone amounts to ignoring the cause to model only the consequence.

Faced with this observation, two paths are possible:

- Continue with the time series approach, by adding information (sentiment, on chain, ...) to attempt to capture the missing factors.
- Or change the problem statement. If the "When" (timing) is unpredictable, then the "What" (asset selection) becomes the critical variable.

For reasons explained in the article, this second solution was retained. As my objective remains to train myself on Web3.0 via Data Science, I did not want to confine myself to an optimization race on a signal that is too noisy.

Rather than focusing on the output variable (price), the subsequent efforts were directed towards the input variables. The idea is to use the different digital traces that projects naturally leave during their creation (code, documentation, on-chain activity).

We have thus set up a new blockchain project evaluation project, to continue tracking the evolutions experienced by the blockchain field. These traces take several forms, addressed with a methodical approach based on:

- A rigorous bibliographic analysis to identify relevant metrics.
- An iterative construction (CI/CD type) to validate data before modeling.

A new roadmap has been established to document this work, structured around five analysis pillars (Team, Technology, Tokenomics, Documentation, Community), the details of which are outlined in the remainder of this logbook.

4. Conclusion: A False Hypothesis, A Validated Approach

This first step is finished. Although it may seem to end in failure since the tested hypothesis was proven false, and that therefore a machine learning model based solely on past prices (I insist on this part) appears unfeasible. But the initial objective, namely, to use this project to start exploring the cryptoverse, is a success.

I discovered a plethora of resources that will be useful to me in the rest of the project. They take the form of diverse data sources, guides, articles, and scientific publications that allowed me to reflect on what is the next step of my journey.

So, without transition, direction the sequel: namely the articles in long version, the published code.